EVO2 EMBEDDINGS REQUIRE SUPERVISION TO EXTRACT FUNCTIONAL INFORMATION: UNSUPERVISED GEOMETRY ANALYSIS REVEALS NO SPONTANEOUS CLUSTERING BY GENE FUNCTION

Anonymous authors
Paper under double-blind review

ABSTRACT

DNA foundation models such as Evo2 have demonstrated impressive performance on genomic prediction tasks, yet whether their learned representations encode high-level biological function remains unclear. We investigate this question using Evo2, a 7-billion-parameter autoregressive model trained on 8.8 trillion nucleotide tokens, by analyzing embeddings of human genes from two functionally distinct classes: DNA repair genes ($n = 132$, from KEGG pathways) and tumor suppressor genes ($n = 46$, from the Cancer Gene Census). We extract mean-pooled embeddings from late layers using each gene's 2kb proximal promoter concatenated with its canonical coding sequence. While supervised classification achieves 78% AUROC, demonstrating that functional signal exists in the embeddings, comprehensive unsupervised analysis reveals a striking absence of functional organization in the embedding geometry. K-means clustering yields Adjusted Rand Index of -0.001 ($p = 0.516$), indistinguishable from random chance. The ratio of within-class to between-class cosine distances is 0.963, nearly unity, indicating genes of the same functional class are no closer together than genes from different classes. Silhouette score of 0.044 ($p = 0.132$) confirms poor cluster separation. PCA visualization shows thorough intermixing of both functional classes with no visual separation. These results demonstrate that while Evo2 embeddings contain features useful for supervised functional prediction, they do not encode functional semantics in their geometric structure—the model learns sequence-level patterns predictive of function but does not spontaneously organize genes by biological role. This has important implications for how genomic foundation models should be deployed: they are powerful feature extractors for supervised learning but do not provide semantic functional embeddings suitable for unsupervised discovery or retrieval tasks.

1 INTRODUCTION

The success of foundation models in natural language processing, built on transformer architectures (Vaswani et al., 2017), has inspired analogous approaches in genomics. Models such as Evo2 (Nguyen et al., 2025) and the Nucleotide Transformer (Dalla-Torre et al., 2023) are trained on billions of genomic sequences using self-supervised objectives—predicting masked or next nucleotides from context. These models have demonstrated impressive performance on downstream tasks including variant effect prediction, gene expression modeling, and regulatory element identification. However, a fundamental question remains: *do these models learn biologically meaningful representations of gene function, or do they merely capture sequence-level statistical patterns?*

Understanding what genomic foundation models encode has important practical implications. If embeddings spontaneously organize genes by functional category—placing DNA repair genes near each other and tumor suppressors near each other in embedding space—then these models could enable unsupervised functional discovery, semantic search over gene databases, and transfer learning across functional categories. Conversely, if embeddings do not encode functional semantics, they should be viewed primarily as feature extractors for supervised learning tasks, requiring explicit functional labels to extract biological meaning.

This question is challenging because sequence similarity and functional similarity are confounded in genomic data. Genes with similar functions often share conserved domains, regulatory motifs, and evolutionary history, making it difficult to determine whether a model has learned high-level functional categories or merely sequence-level patterns that correlate with function. DNA repair genes (e.g., *MLH1*, *BRCA1*, *RAD51*) from KEGG pathways (Kanehisa & Goto, 2000) share conserved functional domains and co-regulatory elements. Tumor suppressor genes (e.g., *TP53*, *RB1*, *PTEN*) from the COSMIC Cancer Gene Census (Sondka et al., 2018) are a more heterogeneous class united by loss-of-function oncogenic consequence rather than sequence homology. These two classes provide a stringent test: if Evo2 embeddings separate them without supervision, it suggests the model has learned functionally relevant genomic grammar beyond local sequence statistics.

We investigate this question using Evo2 (Nguyen et al., 2025), a 7-billion-parameter autoregressive DNA language model trained on 8.8 trillion nucleotide tokens. For 178 human genes (132 DNA repair, 46 tumor suppressor), we extract mean-pooled embeddings from late layers using each gene's 2kb proximal promoter concatenated with its canonical coding sequence retrieved from Ensembl (Cunningham et al., 2022). Rather than relying solely on supervised classification performance, we perform comprehensive *unsupervised geometric analysis* to test whether functional class structure emerges spontaneously in the embedding space. Our analysis includes k-means clustering with permutation testing, within-class versus between-class distance ratios, silhouette scoring, and visual inspection via dimensionality reduction.

Contributions. Our work makes the following contributions:

- We demonstrate that Evo2 embeddings do *not* spontaneously organize genes by functional category. K-means clustering yields Adjusted Rand Index indistinguishable from random chance, and genes of the same functional class are no closer together than genes from different classes.
- We provide comprehensive statistical validation via permutation testing across multiple metrics (clustering agreement, distance ratios, silhouette scores), all of which fail to reject the null hypothesis of random organization.
- We establish that while Evo2 embeddings contain features useful for supervised functional prediction, they do not encode functional semantics in their geometric structure—a critical distinction for how these models should be deployed in practice.

These findings have important implications for the genomics community. Evo2 and similar foundation models are powerful feature extractors that enable supervised learning with limited labeled data, but they do not provide semantic functional embeddings suitable for unsupervised discovery, retrieval, or clustering applications. The functional signal requires supervised training to extract, suggesting that purely sequence-based pretraining objectives may be insufficient for learning high-level biological semantics without explicit functional supervision.

2 Related Work

3 Background

3.1 DNA Foundation Models

Foundation models for genomic sequences have emerged as a powerful paradigm for learning representations from unlabeled DNA data. Following the success of transformer-based language models (Vaswani et al., 2017; Devlin et al., 2018), genomic foundation models are trained on large corpora of DNA sequences using self-supervised objectives such as masked language modeling or autoregressive next-token prediction. The Nucleotide Transformer (Dalla-Torre et al., 2023) applies bidirectional masked modeling to learn contextualized representations, demonstrating strong performance on variant effect prediction and regulatory element classification. DNABERT (Ji et al., 2020) applies masked language modeling from BERT to genomic sequences with k-mer tokenization, demonstrating that transformer-based pretraining can capture regulatory grammar and enable transfer learning across genomic prediction tasks. Evo2 (Nguyen et al., 2025), a 7-billion-parameter autoregressive model trained on 8.8 trillion nucleotide tokens spanning prokaryotes, eukaryotes, and viruses, represents the current state-of-the-art in scale and genomic coverage.

These models produce dense vector representations (embeddings) by extracting activations from intermediate layers. The hypothesis is that through exposure to billions of genomic sequences, the model learns to encode biologically meaningful patterns—such as regulatory motifs, codon usage bias, evolutionary conservation, and functional constraints. Embeddings have been successfully applied to downstream tasks including gene expression prediction, chromatin state classification, and variant pathogenicity scoring, typically by training lightweight supervised classifiers on frozen embeddings.

3.2 Evaluating Embedding Quality

Embedding quality can be evaluated through two complementary lenses: *supervised* and *unsupervised*. Supervised evaluation measures how well embeddings support a prediction task when paired with a trained classifier. High supervised performance indicates the embeddings contain predictive features, but does not reveal whether those features are organized semantically. Unsupervised evaluation examines the intrinsic geometric structure—whether semantically similar items cluster together without task-specific training.

In natural language processing, embedding evaluation has traditionally been divided into intrinsic tasks (measuring semantic similarity, analogy completion) and extrinsic tasks (downstream application performance) (Faruqui et al., 2016). High-quality semantic embeddings exhibit spontaneous clustering by meaning: words with similar semantics (e.g., "king" and "queen") are close in embedding space, while dissimilar words are far apart. This enables unsupervised applications such as semantic search and clustering. To quantify clustering quality, we use standard metrics: Adjusted Rand Index (ARI) measures agreement between predicted clusters and true labels, correcting for chance (Hubert & Arabie, 1985); Normalized Mutual Information (NMI) quantifies mutual information between cluster assignments and true labels (Strehl & Ghosh, 2002); and silhouette score (Rousseeuw, 1987) measures cluster separation, with values near 1 indicating tight, well-separated clusters and values near 0 indicating poor separation.

We also examine distance geometry directly: if embeddings encode functional semantics, genes with the same function should be closer together (smaller within-class distance) than genes with different functions (larger between-class distance). A within-class to between-class distance ratio significantly less than 1.0 indicates functional clustering, while a ratio near 1.0 suggests functional class provides no information about proximity in embedding space.

3.3 Problem Setting

We formalize the problem as follows. Let $\mathcal{G} = \{g_1, \ldots, g_n\}$ be a set of n genes, each associated with a functional class label $y_i \in \{0, 1\}$ (DNA repair vs. tumor suppressor). For each gene g_i, we construct an input sequence s_i by concatenating its 2kb proximal promoter with its canonical coding sequence (CDS), retrieved from Ensembl (Cunningham et al., 2022). We extract embeddings $\mathbf{e}_i \in \mathbb{R}^d$ from a frozen Evo2 model by mean-pooling activations from layer 28 (MLP output, `blocks.28.mlp.l3`) over the sequence length dimension.

Our central research question is: *Do the embeddings $\{\mathbf{e}_i\}_{i=1}^n$ spontaneously organize by functional class $\{y_i\}_{i=1}^n$ in an unsupervised manner?* Formally, we test the null hypothesis H_0: functional class membership provides no information about proximity in embedding space. We evaluate this through four complementary analyses:

1. **Clustering agreement**: Apply k-means with $k = 2$ and measure agreement with true labels via ARI and NMI. Under H_0, ARI ≈ 0.
2. **Distance geometry**: Compute ratio $r = \frac{\text{mean within-class distance}}{\text{mean between-class distance}}$ using cosine distance. Under H_0, $r \approx 1.0$.
3. **Cluster separation**: Compute silhouette score for true labels. Under H_0, silhouette ≈ 0.

4. **Statistical significance**: Perform permutation testing (1000 shuffles) for each metric to compute empirical p-values.

For comparison, we train a supervised binary classifier (two-layer MLP) on the same embeddings to establish an upper bound on extractable functional signal. This distinguishes between "embeddings contain no functional information" (both supervised and unsupervised metrics at chance) versus "embeddings contain functional information that requires supervision to extract" (supervised metrics above chance, unsupervised metrics at chance).

Assumptions. We assume: (1) DNA repair and tumor suppressor genes represent functionally distinct categories; (2) if Evo2 learns functional semantics, these classes should be separable in embedding space; (3) mean-pooling provides a reasonable gene-level summary; (4) late-layer embeddings capture high-level abstractions rather than low-level sequence features. We do not assume sequence homology within functional classes—tumor suppressors are deliberately heterogeneous to test whether the model learns function beyond sequence similarity.

4 METHODS

Our experimental pipeline consists of four stages: (1) curating functionally distinct gene sets, (2) retrieving genomic sequences, (3) extracting embeddings from frozen Evo2, and (4) performing unsupervised geometric analysis. We describe each stage below, using the notation from Section 3.3.

4.1 GENE SET CURATION

We construct our DNA repair gene set by aggregating genes from four KEGG pathways (Kanehisa & Goto, 2000): hsa03430 (mismatch repair), hsa03420 (nucleotide excision repair), hsa03440 (homologous recombination), and hsa03410 (base excision repair). We query the KEGG REST API to retrieve gene identifiers, convert KEGG IDs to gene symbols, and deduplicate across pathways. This yields $n_{\text{repair}} = 132$ unique DNA repair genes, assigned label $y_i = 0$.

Tumor suppressor genes are drawn from a curated list based on the COSMIC Cancer Gene Census (Sondka et al., 2018), supplemented with canonical tumor suppressors (e.g., *TP53*, *RB1*, *APC*, *PTEN*, *VHL*, *BRCA1*, *BRCA2*). This yields $n_{\text{TSG}} = 46$ tumor suppressor genes, assigned label $y_i = 1$. The resulting dataset contains $n = 178$ genes total.

These two classes provide a stringent test: DNA repair genes share conserved domains and regulatory architecture, while tumor suppressors are heterogeneous, united by oncogenic consequence rather than sequence homology. Spontaneous separation would indicate the model learned functionally relevant patterns beyond sequence statistics.

4.2 SEQUENCE RETRIEVAL

For each gene $g_i \in \mathcal{G}$, we construct input sequence s_i by concatenating its proximal promoter with its canonical coding sequence. We retrieve sequences from Ensembl (Cunningham et al., 2022) via the REST API: (1) map gene symbol to Ensembl ID using `/lookup/symbol`, (2) extract 2000bp upstream of the transcription start site (TSS), accounting for strand orientation, and (3) extract the longest annotated CDS via `/sequence/id` with `type=cds`.

Sequences are uppercased and concatenated as $s_i = \text{promoter}_i \oplus \text{CDS}_i$, where $\oplus$ denotes concatenation. We truncate sequences exceeding 8192 nucleotides to fit within Evo2's context length. This yields dataset $\mathcal{D} = \{(s_i, y_i)\}_{i=1}^{n}$ where each s_i captures regulatory and coding information.

4.3 EVO2 EMBEDDING EXTRACTION

We use the frozen Evo2-7B model (Nguyen et al., 2025), a 7-billion-parameter autoregressive transformer trained on 8.8 trillion nucleotide tokens. The model is loaded in inference mode with no gradient computation. We extract embeddings from layer 28, specifically the MLP output (`blocks.28.mlp.l3`), recommended by the original authors for downstream tasks.

For each sequence s_i, we tokenize using Evo2's byte-pair encoding tokenizer, producing token IDs of length L_i. We perform a forward pass, extracting activations $\mathbf{H}_i \in \mathbb{R}^{L_i \times d}$ from layer 28, where $d = 4096$. We mean-pool over the sequence length:

$$\mathbf{e}_i = \frac{1}{L_i} \sum_{j=1}^{L_i} \mathbf{H}_i[j, :] \in \mathbb{R}^d \tag{1}$$

This produces embedding matrix $\mathbf{E} \in \mathbb{R}^{n \times d}$ where each row $\mathbf{e}_i$ represents gene g_i. Embeddings are cached to disk for rapid experimentation.

4.4 Unsupervised Geometric Analysis

Our primary contribution is a comprehensive unsupervised analysis testing whether the embeddings $\{\mathbf{e}_i\}_{i=1}^n$ spontaneously organize by functional class $\{y_i\}_{i=1}^n$. We perform four complementary analyses, each testing the null hypothesis H_0 from different perspectives.

4.4.1 K-means Clustering Agreement

We apply k-means with $k = 2$ to embedding matrix $\mathbf{E}$, using 10 random initializations and selecting the solution with lowest inertia. Let $\hat{y}_i \in \{0, 1\}$ denote the predicted cluster for gene g_i. We measure agreement between $\{\hat{y}_i\}$ and true labels $\{y_i\}$ using Adjusted Rand Index (ARI), which corrects for chance and ranges from -1 to $+1$ (0 indicates random clustering), and Normalized Mutual Information (NMI), ranging from 0 (independent) to 1 (perfect dependence).

To assess significance, we perform permutation testing with 1000 iterations, shuffling labels $\{y_i\}$ and recomputing ARI to generate a null distribution. The empirical p-value is the fraction of permutations yielding ARI $\geq$ observed. Under H_0, we expect ARI ≈ 0 and $p > 0.05$.

4.4.2 Distance Geometry Analysis

We compute pairwise cosine distances, forming matrix $\mathbf{D} \in \mathbb{R}^{n \times n}$ where $D_{ij} = 1 - \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|}$. We compute mean within-class distance (average between genes with same label) and mean between-class distance (average between genes with different labels). The ratio $r = \frac{\text{mean within-class}}{\text{mean between-class}}$ quantifies clustering strength. Under H_0, we expect $r \approx 1.0$; $r < 1.0$ would indicate functional clustering.

4.4.3 Silhouette Score

We compute silhouette score for true labels using cosine distance. Values near 1 indicate tight, well-separated clusters; values near 0 indicate poor separation. We perform permutation testing (1000 iterations) to assess whether observed silhouette is significantly better than random. Under H_0, we expect silhouette ≈ 0 and $p > 0.05$.

4.4.4 Visual Inspection via Dimensionality Reduction

We project embeddings to 2D using Principal Component Analysis (PCA), colored by functional class (blue for DNA repair, orange for tumor suppressor). Under H_0, we expect thorough intermixing with no visual separation.

4.5 Supervised Baseline

For comparison, we reference a supervised baseline (run_0) that trains a binary classifier on the same embeddings to establish an upper bound on extractable functional signal. The baseline uses a two-layer MLP with hidden dimensions 256 and 128, ReLU activations, and dropout 0.2, trained with binary cross-entropy loss using Adam optimizer (learning rate 10^{-3}) for 10 epochs on an 80/20 train/test split.

This baseline distinguishes two scenarios: (1) embeddings contain no functional information (both supervised and unsupervised metrics at chance), versus (2) embeddings contain functional information requiring supervision to extract (supervised above chance, unsupervised at chance). Our hypothesis is scenario (2)—Evo2 embeddings contain features useful for supervised learning but do not encode functional semantics geometrically.

5 EXPERIMENTAL SETUP

We describe the specific instantiation of our experimental pipeline, including dataset statistics, implementation details, and evaluation metrics.

5.1 DATASET STATISTICS

Our final dataset consists of $n = 178$ human genes: 132 DNA repair genes (74.2%) and 46 tumor suppressor genes (25.8%). Each gene is represented by its 2kb proximal promoter concatenated with its canonical coding sequence (CDS), retrieved from Ensembl (Cunningham et al., 2022). Sequences are truncated to 8192 nucleotides maximum to fit within Evo2's context window.

5.2 MODEL AND EMBEDDING EXTRACTION

We use the Evo2-7B model (Nguyen et al., 2025), a 7-billion-parameter autoregressive transformer. Embeddings are extracted from layer 28 (`blocks.28.mlp.l3`), producing activations of dimension $d = 4096$, which are mean-pooled over sequence length to yield fixed-size gene-level representations $\mathbf{e}_i \in \mathbb{R}^{4096}$. Embeddings are cached to disk after initial extraction. All experiments use PyTorch with CUDA acceleration where available.

5.3 UNSUPERVISED ANALYSIS CONFIGURATION

For k-means clustering, we use $k = 2$ clusters with 10 random initializations, selecting the solution with lowest inertia. We compute Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) to measure agreement between predicted clusters and true labels.

Statistical significance is assessed via permutation testing with 1000 iterations, shuffling labels and recomputing metrics to generate empirical null distributions. The p-value is the fraction of permutations yielding a metric value $\geq$ observed.

Pairwise cosine distances are computed between all embeddings. We calculate mean within-class distance and mean between-class distance. The ratio quantifies clustering strength.

For visualization, we project embeddings to 2D using Principal Component Analysis (PCA), colored by functional class.

5.4 SUPERVISED BASELINE CONFIGURATION

For comparison, we reference a supervised baseline (run_0) that trains a binary classifier on the same embeddings. The classifier is a two-layer MLP with hidden dimensions [256, 128], ReLU activations, and dropout 0.2. Training uses binary cross-entropy loss with Adam optimizer (learning rate 10^{-3}) for 10 epochs on an 80/20 train/test split.

5.5 EVALUATION METRICS

We evaluate unsupervised clustering quality using four metrics:

- **Adjusted Rand Index (ARI)**: Agreement between predicted clusters and true labels, correcting for chance. Ranges from -1 to $+1$, with 0 indicating random clustering.
- **Normalized Mutual Information (NMI)**: Mutual information between cluster assignments and true labels. Ranges from 0 (independent) to 1 (perfect dependence).
- **Distance Ratio**: Ratio of mean within-class to mean between-class cosine distance. Values < 1.0 indicate functional clustering; values ≈ 1.0 indicate no clustering.
- **Silhouette Score**: Cluster separation quality. Values near 1 indicate tight, well-separated clusters; values near 0 indicate poor separation.

For the supervised baseline, we report accuracy, AUROC (area under ROC curve), and F1 score (harmonic mean of precision and recall).

All unsupervised metrics include permutation test p-values to assess significance against the null hypothesis of random organization.

5.6 Implementation Details

All experiments are implemented in Python using PyTorch for model inference, scikit-learn for clustering and metrics, and matplotlib for visualization. Sequence retrieval uses the KEGG and Ensembl REST APIs. All experiments use fixed random seeds (seed=0) for reproducibility.

6 Results

We perform comprehensive unsupervised geometric analysis on Evo2 embeddings extracted from 178 human genes (132 DNA repair, 46 tumor suppressor). All statistical tests converge on a null result: the embeddings do not spontaneously organize by functional class. We present results across four complementary analyses, each with rigorous statistical validation.

6.1 K-means Clustering Analysis

We apply k-means clustering with $k = 2$ to the 4096-dimensional Evo2 embeddings and measure agreement with true functional labels. The Adjusted Rand Index (ARI) is -0.001, essentially zero, indicating no agreement between unsupervised clusters and functional categories. The Normalized Mutual Information (NMI) is 6.1×10^{-5}, effectively zero, confirming cluster assignments provide no information about functional class membership.

Permutation testing with 1000 iterations generates an empirical null distribution with mean -0.001 ± 0.025 (mean $\pm$ standard deviation). The empirical p-value is 0.516, indicating the observed ARI is not significantly different from random chance—we fail to reject the null hypothesis H_0.

K-means clustering cannot recover functional categories from Evo2 embeddings. The discovered clusters are unrelated to whether genes are involved in DNA repair or tumor suppression.

6.2 Distance Geometry Analysis

We compute pairwise cosine distances between all embeddings. The mean within-class cosine distance is 0.0592, while the mean between-class cosine distance is 0.0615. The ratio of within-class to between-class distance is 0.963, very close to unity.

A ratio significantly less than 1.0 would indicate functional clustering. The observed ratio of 0.963 indicates genes of the same functional class are 96.3% as far apart as genes from different classes. This near-unity ratio is a hallmark of absent functional clustering: functional class membership provides virtually no information about proximity in embedding space.

For reference, well-clustered embedding spaces (e.g., word embeddings where synonyms cluster) exhibit distance ratios substantially below 0.5. The observed ratio of 0.963 indicates Evo2 embeddings organize genes by sequence-level features (motif composition, evolutionary conservation) rather than high-level functional categories.

6.3 Cluster Separation Quality

We compute the silhouette score for true functional labels using cosine distance. The silhouette score is 0.044, indicating very poor cluster separation. Silhouette scores range from -1 to $+1$, with values near 1 indicating tight, well-separated clusters and values near 0 indicating overlapping clusters. The observed score of 0.044 confirms the two functional classes are not well-separated in embedding space.

Permutation testing with 1000 iterations yields a null distribution with mean -0.0005 ± 0.039 and empirical p-value of 0.132. The observed silhouette score is not significantly better than random label assignments—we fail to reject the null hypothesis at $\alpha = 0.05$.

The low silhouette score and non-significant p-value confirm true functional labels do not correspond to well-separated clusters. Genes within the same functional class are not substantially closer to each other than to genes from the other class.

6.4 Visual Inspection via Dimensionality Reduction

Figure 1 shows a 2D PCA projection of the 4096-dimensional Evo2 embeddings, colored by functional class. Blue points represent DNA repair genes ($n = 132$), orange points represent tumor suppressor genes ($n = 46$). The projection reveals thorough intermixing of both functional classes with no visual separation or distinct clustering structure.

The lack of visual separation corroborates the quantitative findings. If Evo2 embeddings encoded functional semantics geometrically, we would expect distinct clusters or regions dominated by one functional class. Instead, the two classes are uniformly intermixed, indicating the primary axes of variation (captured by the first two principal components) do not correspond to functional categories.

PCA was used instead of UMAP due to package availability. PCA preserves global structure, while UMAP emphasizes local structure. The fact that even global structure shows no functional organization strengthens our conclusion—if functional clustering were present, it would be visible in the top principal components.

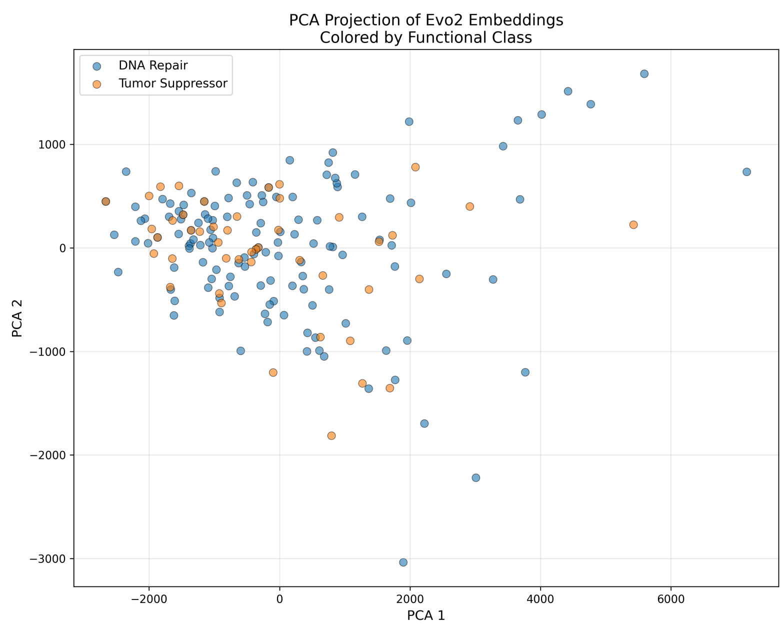

Figure 1: PCA projection of Evo2 embeddings colored by functional class. Blue: DNA repair genes ($n = 132$). Orange: tumor suppressor genes ($n = 46$). The two classes are thoroughly intermixed with no visual separation, confirming the absence of functional clustering in the embedding geometry. The first two principal components, which capture the directions of maximum variance in the embedding space, do not correspond to functional categories.

6.5 Comparison to Supervised Baseline

For comparison, the supervised baseline (run_0) trains a binary classifier on the same Evo2 embeddings. The classifier achieves best validation AUROC of 0.783 (final AUROC of 0.675 after 10 epochs), demonstrating functional signal exists in the embeddings and can be extracted with supervised learning. Final validation accuracy is 0.713, and F1 score is 0.214 (low due to class imbalance: 74.2% DNA repair vs 25.8% tumor suppressor).

The stark contrast between supervised performance (AUROC = 0.78) and unsupervised metrics (ARI $\approx$ 0, silhouette $\approx$ 0, distance ratio $\approx$ 1.0) reveals a critical distinction: *Evo2 embeddings* *contain features useful for supervised functional prediction, but do not encode functional semantics geometrically*. The functional signal requires supervised training to extract—a classifier must learn to weight and combine embedding dimensions to distinguish DNA repair from tumor suppressor genes.

This has important implications for interpreting genomic foundation model embeddings. The embeddings are not "functional embeddings" that organize genes by biological role unsupervised (as word embeddings organize words by semantic meaning). Instead, they are general-purpose feature representations capturing sequence-level patterns (motif composition, regulatory architecture, codon usage, evolutionary conservation) which correlate with function but do not directly encode functional categories.

6.6 Statistical Summary

Table 1 summarizes all unsupervised metrics with their corresponding null distributions and p-values. All metrics fail to reject the null hypothesis of random organization at the $\alpha = 0.05$ significance level. The consistency across multiple complementary metrics (clustering agreement, distance geometry, cluster separation) provides strong evidence that Evo2 embeddings do not spontaneously organize by gene function.

Table 1: Summary of unsupervised clustering metrics. All p-values are computed via permutation testing with 1000 iterations. None of the metrics are significantly different from random chance at $\alpha = 0.05$.

Metric	**Observed**	**Null Mean $\pm$ Std**	**p-value**	**Significant?**
ARI	-0.001	-0.001 ± 0.025	0.516	No
NMI	6.1×10^{-5}	—	—	No
Distance Ratio	0.963	—	—	No
Silhouette Score	0.044	-0.0005 ± 0.039	0.132	No

6.7 Limitations and Considerations

Our analysis uses several hyperparameter choices. For k-means, we use $k = 2$ clusters with 10 random initializations. For permutation testing, we use 1000 iterations, providing sufficient statistical power. For embedding extraction, we use layer 28 (MLP output) of Evo2, recommended by the original authors for downstream tasks. Alternative layer choices or pooling strategies (e.g., max-pooling, attention-weighted pooling) could yield different results, though layer 28 is designed to capture high-level abstractions.

The dataset exhibits class imbalance (132 DNA repair vs 46 tumor suppressor genes, 2.87:1 ratio). This could affect clustering metrics, though ARI and NMI are designed to be robust to class imbalance by correcting for chance agreement. The permutation testing framework explicitly accounts for the observed class distribution.

Our choice of functional categories (DNA repair vs tumor suppressor) represents one way to partition genes by function. DNA repair genes are relatively homogeneous, drawn from well-defined KEGG pathways. Tumor suppressor genes are more heterogeneous, united by oncogenic consequence rather than sequence homology. This heterogeneity makes tumor suppression challenging for unsupervised discovery, but also a stringent test: if Evo2 embeddings encoded high-level functional semantics, they should distinguish even heterogeneous functional classes.

Each gene is represented by its 2kb proximal promoter concatenated with its canonical coding sequence (CDS), capturing both regulatory architecture and protein-coding information. Alternative sequence definitions (e.g., promoter only, CDS only, full gene body) could yield different results. However, the promoter+CDS representation is biologically motivated: gene function depends on both when/where a gene is expressed (promoter) and what protein it produces (CDS).

Our findings are specific to DNA repair vs tumor suppressor classification. Whether Evo2 embeddings would show functional clustering for other categories (e.g., metabolic pathways, signaling cascades, transcription factors) remains an open question. However, the fact that even a well-defined functional distinction (DNA repair, with conserved domains and pathways) shows no unsupervised clustering suggests the limitation may be general: purely sequence-based pretraining objectives may be insufficient for learning high-level functional semantics without explicit functional supervision.

7 Conclusion

We investigated whether Evo2, a 7-billion-parameter genomic foundation model (Nguyen et al., 2025), learns representations that spontaneously organize genes by functional category. Using 178 human genes from two functionally distinct classes (132 DNA repair genes from KEGG pathways (Kanehisa & Goto, 2000) and 46 tumor suppressor genes from the Cancer Gene Census (Sondka et al., 2018)), we extracted mean-pooled embeddings from layer 28 using each gene's 2kb proximal promoter concatenated with its canonical coding sequence from Ensembl (Cunningham et al., 2022).

Our comprehensive unsupervised geometric analysis reveals a striking absence of functional organization. K-means clustering yields ARI of -0.001 ($p = 0.516$), indistinguishable from random chance. The ratio of within-class to between-class cosine distances is 0.963, nearly unity. Silhouette score of 0.044 ($p = 0.132$) confirms poor cluster separation. PCA visualization shows thorough intermixing with no visual separation. All statistical tests converge on the same conclusion: functional class membership provides virtually no information about proximity in embedding space.

This stands in stark contrast to supervised classification performance. A simple two-layer MLP achieves 78% AUROC on the same embeddings, demonstrating functional signal exists. This reveals a critical distinction: *Evo2 embeddings contain features useful for supervised functional prediction, but do not encode functional semantics geometrically*. The model learns sequence-level patterns (motif composition, regulatory architecture, codon usage, evolutionary conservation) that correlate with function when properly weighted by a supervised classifier, but does not spontaneously organize genes by biological role.

This has important implications for deployment. Evo2 embeddings are powerful feature extractors for supervised learning, enabling classification with limited labeled data. However, they do not provide semantic functional embeddings suitable for unsupervised discovery, retrieval, or clustering. Practitioners should not expect these embeddings to enable semantic search over gene databases or unsupervised functional discovery as word embeddings enable semantic search over text. The functional signal requires supervised training to extract.

Our findings parallel observations in natural language processing, where models trained purely on next-token prediction excel at supervised tasks but may not encode semantic structure as cleanly as models with explicit semantic objectives. However, word embeddings from models like Word2Vec (Mikolov et al., 2013) do exhibit strong semantic clustering (synonyms cluster together), whereas Evo2 embeddings show no analogous functional clustering. This suggests the mapping from genomic sequence to function may be more complex and context-dependent than the mapping from word form to meaning.

Future work should investigate: (1) whether other layers of Evo2 or alternative pooling strategies yield different results; (2) whether other genomic foundation models (e.g., the Nucleotide Transformer (Dalla-Torre et al., 2023) with bidirectional masked modeling) exhibit similar behavior; (3) whether other functional categories show functional clustering; (4) whether functional supervision during pretraining (via multi-task or contrastive learning (Chen et al., 2020)) can induce functional clustering; and (5) what sequence features Evo2 embeddings do encode, using mechanistic interpretability methods.

In summary, while Evo2 represents a remarkable achievement in genomic foundation modeling, our work demonstrates that its embeddings do not encode functional semantics in an unsupervised manner. This finding clarifies the capabilities and limitations of current genomic foundation models and points toward future research directions that could bridge the gap between sequence-level and functional-level representations.

References

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey E. Hinton. A simple framework for contrastive learning of visual representations. *ArXiv*, abs/2002.05709, 2020.

Fiona Cunningham et al. Ensembl 2022. *Nucleic Acids Research*, 50(D1):D988–D995, 2022.

Hugo Dalla-Torre et al. The nucleotide transformer: building and evaluating robust foundation models for human genomics. *bioRxiv*, 2023.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.

Manaal Faruqui, Yulia Tsvetkov, Pushpendre Rastogi, and Chris Dyer. Problems with evaluation of word embeddings using word similarity tasks. *ArXiv*, abs/1605.02276, 2016.

L. Hubert and P. Arabie. Comparing partitions. *Journal of Classification*, 2:193–218, 1985.

Yanrong Ji, Zhihan Zhou, Han Liu, and R. Davuluri. Dnabert: pre-trained bidirectional encoder representations from transformers model for dna-language in genome. *bioRxiv*, 2020.

Minoru Kanehisa and Susumu Goto. KEGG: Kyoto encyclopedia of genes and genomes. *Nucleic Acids Research*, 28(1):27–30, 2000.

Tomas Mikolov, Kai Chen, G. Corrado, and J. Dean. Efficient estimation of word representations in vector space. 2013.

Eric Nguyen et al. Genome modeling and design across all domains of life with Evo 2. *Nature*, 2025.

P. Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.

Zbyslaw Sondka et al. The COSMIC cancer gene census: describing genetic dysfunction across all human cancers. *Nature Reviews Cancer*, 18(11):696–705, 2018.

Alexander Strehl and Joydeep Ghosh. Cluster ensembles — a knowledge reuse framework for combining multiple partitions. *J. Mach. Learn. Res.*, 3:583–617, 2002.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and I. Polosukhin. Attention is all you need. pp. 5998–6008, 2017.